10 SQL QUERIES FOR HEALTH DATA ANALYSIS

Practical SQL examples demonstrating health dataset querying, patient trends, and public health metrics.

1. Total Patient Admissions by Facility

```
SELECT facility_name, COUNT(patient_id)
AS total_admissions
FROM admissions
GROUP BY facility_name
ORDER BY total_admissions DESC;
```

Aggregates patient volume across healthcare centers to identify resource utilization bottlenecks.

2. Average Length of Stay (LOS) by Diagnosis

```
SELECT diagnosis_code,
  ROUND(AVG(DATEDIFF(discharge_date,
admission_date)), 1) AS avg_los_days
FROM patient_encounters
GROUP BY diagnosis_code;
```

Calculates average recovery time per diagnosis category to support hospital planning.

3. 30-Day Hospital Readmission Rate

```
SELECT
  ROUND(100.0 * SUM(CASE WHEN
readmitted_30_days = 1 THEN 1 ELSE 0
END) / COUNT(*), 2) AS readmission_rate
FROM patient_outcomes;
```

Measures healthcare quality and post-discharge complication frequency.

4. Identifying Outlier Vitals (Hypertension Screen)

```
SELECT patient_id, recorded_date,
systolic_bp, diastolic_bp
FROM vital_signs
WHERE systolic_bp > 140 OR diastolic_bp
> 90
ORDER BY systolic_bp DESC;
```

Filters patients presenting with elevated blood pressure thresholds for clinical review.

5. Monthly Disease Incident Trends

```
SELECT DATE_FORMAT(diagnosis_date, '%Y-
%m') AS month_year,
  COUNT(patient_id) AS new_cases
FROM disease_registry
GROUP BY month_year
ORDER BY month_year ASC;
```

Tracks epidemiological patterns and seasonal disease outbreaks over time.

6. Demographic Breakdown of Vaccinated Patients

```
SELECT age_group, gender, COUNT(patient_id) AS
vaccinated_count
FROM vaccinations
WHERE vaccine_status = 'Fully Vaccinated'
GROUP BY age_group, gender;
```

Evaluates immunization coverage across specific age brackets and gender groups.

7. Stock Level Audit for Essential Medicines

```
SELECT item_name, stock_quantity,
reorder_threshold
FROM pharmacy_inventory
WHERE stock_quantity <= reorder_threshold
ORDER BY stock_quantity ASC;
```

Flags critical medical supplies nearing depletion to prevent stockouts.

8. Average Wait Times in Emergency Triage

```
SELECT triage_category,

ROUND(AVG(TIME_TO_SEC(TIMEDIFF(doctor_seen_time,
arrival_time)) / 60), 1) AS avg_wait_mins
FROM er_visits
GROUP BY triage_category;
```

Monitors emergency response efficiency against priority severity triage tiers.

9. Lab Test Result Turnaround Time

```
SELECT test_type,
  ROUND(AVG(TIMESTAMPDIFF(HOUR,
sample_drawn_time, result_published_time)), 1)
AS avg_turnaround_hours
FROM laboratory_logs
GROUP BY test_type;
```

Analyzes diagnostic processing efficiency across different pathology tests.

10. Pediatric Growth Monitoring (BMI Percentiles)

```
SELECT child_id, age_months, weight_kg,
height_cm,
  ROUND(weight_kg / POWER(height_cm / 100, 2),
2) AS calculated_bmi
FROM child_anthropometry
WHERE age_months <= 60;
```

Computes nutritional indices for child health and growth tracking programs.